

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

# Load Data
df = pd.read_csv("/content/data.csv")
print("--- Data Preview ---")
print(df.head())

# 1. LINE CHART
x = np.linspace(0, 10, 100)
y = np.sin(x)

plt.figure()
plt.plot(x, y, color='blue', label='Sine Wave')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.legend()
plt.show()

# 2. BAR CHART
categories = ['A', 'B', 'C', 'D']
values = [3, 7, 5, 4]

plt.figure()
plt.bar(categories, values, color='orange')
plt.xlabel('Category')
plt.ylabel('Values')
plt.title('Bar Plot')
plt.show()
```

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--- Data Preview ---

	Make	Model	Year	Engine	Fuel Type	Engine HP	\
0	BMW	1 Series M	2011	premium unleaded	(required)	335.0	
1	BMW	1 Series	2011	premium unleaded	(required)	300.0	
2	BMW	1 Series	2011	premium unleaded	(required)	300.0	
3	BMW	1 Series	2011	premium unleaded	(required)	230.0	
4	BMW	1 Series	2011	premium unleaded	(required)	230.0	

	Engine	Cylinders	Transmission	Type	Driven Wheels	Number of Doors	\
0		6.0		MANUAL	rear wheel drive	2.0	
1		6.0		MANUAL	rear wheel drive	2.0	
2		6.0		MANUAL	rear wheel drive	2.0	
3		6.0		MANUAL	rear wheel drive	2.0	
4		6.0		MANUAL	rear wheel drive	2.0	

	Market Category	Vehicle Size	Vehicle Style	\
0	Factory Tuner,Luxury,High-Performance	Compact	Coupe	
1	Luxury,Performance	Compact	Convertible	
2	Luxury,High-Performance	Compact	Coupe	
3	Luxury,Performance	Compact	Coupe	
4	Luxury	Compact	Convertible	

	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	25	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500

3. HISTOGRAM

```
data = np.random.randn(1000)
```

```
plt.figure()
plt.hist(data, bins=20, color='purple', edgecolor='black')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```





